

What use of AI in finance?

ISCF Final Presentation

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Objectives



PROVIDE A STUDY ON THE VARIOUS
APPLICATIONS OF MACHINE LEARNING
IN FINANCE



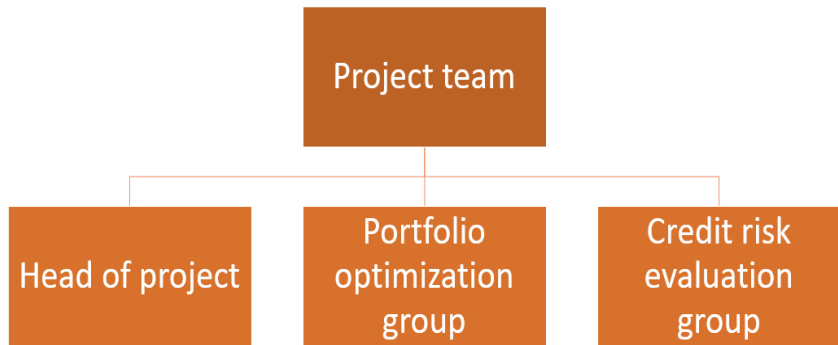
STUDY THE PORTFOLIO MANAGEMENT
PROBLEM AND THE APPLICATION OF
THE SVM & DEEP LEARNING METHODS



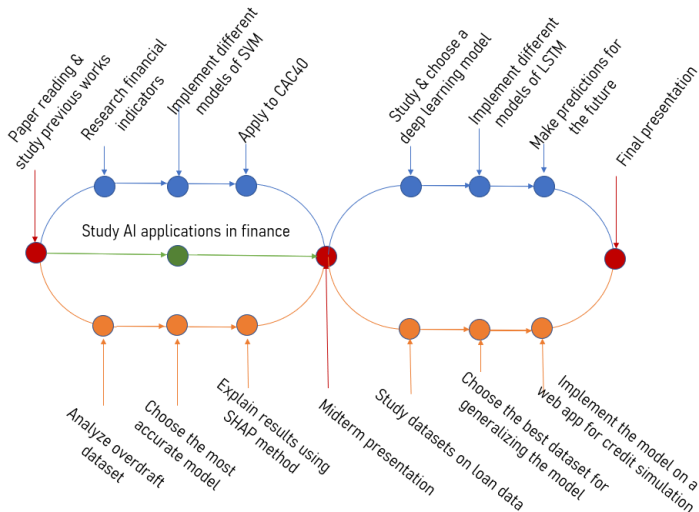
STUDY THE TOOL DEVELOPED FOR THE
APPLICATION OF CREDIT RISK
ASSESSMENT



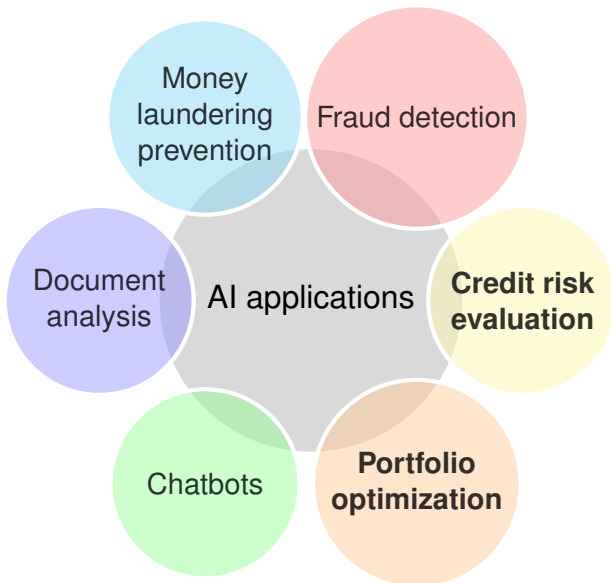
Team organization



Main milestones



AI applications in finance



Portfolio Management

Context

Objective

Provide a tool for investors to support decisions on investing in a specific stock or not, by predicting stock market behaviour.

- Implement machine learning algorithm SVM
- Implement Deep Learning algorithm LSTM
- Make 7-day predictions for market trend with LSTM



CAC40 dataset



- Source: Yahoo! Finance, FCHI ticker
- Contains information from 25/02/1990 up to the current date
- Essentially a time series with 1605 samples, 6 features



Part 1: Support Vector Machine

Predicting stock market changes

2 main theories : Fundamental analysis vs Technical Analysis

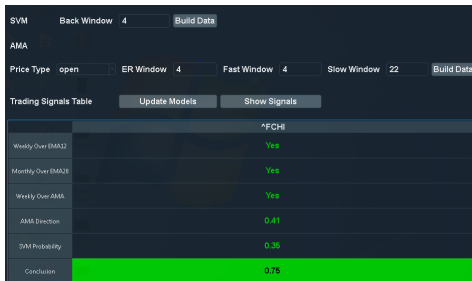
Support Vector Machines(SVM)

SVM is the machine learning algorithm used in classification problems

Our contribution

- Reading and understanding the financial and technical aspects of the project
- Adapted pre-processing steps to conform to our dataset
- Tuned learning parameters to help improve predictions for our scenario

Implementation: Trading Signal



Trading Signals Table	
	*FCHI
Weekly Over EMA12	Yes
Monthly Over EMA26	Yes
Weekly Over AMA	Yes
AMA Direction	0.41
SVM Probability	0.36
Conclusion	0.75

- Five trading signals that perform a weighted average conclusion
- Conclusion reflects the trend of the data (uptrend, downtrend or sideways)
- Backtests show that long-term predictions are more certain

Part 2 : LSTM

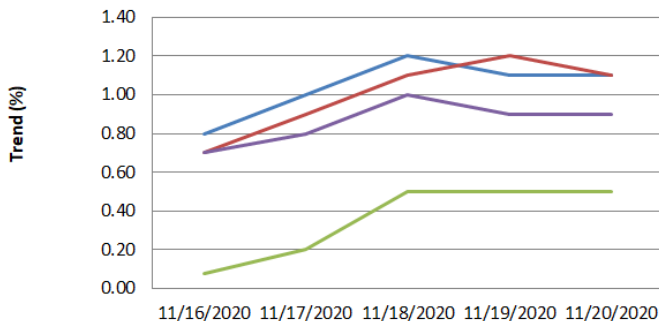
Context

- LSTM is a deep neural network model used for stock prediction
- Contrary to SVM, it is a regression model for the near future predictions
- Training datasets that are used for LSTM
 1. **FCHI** - CAC40 French stock market index
 2. **AEX** - Amsterdam Stock Exchange
 3. **BFX** - BEL20 Belgium Stock Exchange
 4. **GR** - German Stock Market
 5. **IB** - Spanish Stock Market



LSTM results

LSTM 5-days trend



	11/16/2020	11/17/2020	11/18/2020	11/19/2020	11/20/2020
True Values	0.80	1.00	1.20	1.10	1.10
LSTM ADV (FCHI+AEX+BFX)	0.70	0.90	1.10	1.20	1.10
LSTM Vanilla(FCHI)	0.08	0.20	0.50	0.50	0.50
LSTM ADV(FCHI+GR+IB)	0.7	0.8	1	0.9	0.9

Credit risk Evaluation

Context and Methodology

Objective

Automation of the process of approving an overdraft.

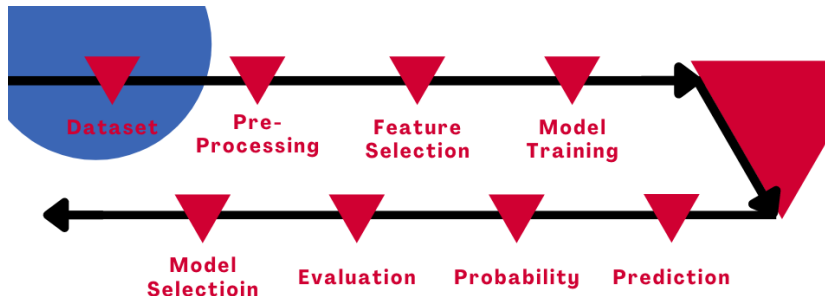


Figure: Schema of methodology



Final model & evaluation

- Balanced Random Forest
- Simple model and similar results to other methods

Method	Specificity	Sensitivity	Accuracy	AUC
BRF	0.785	0.707	0.712	0.746



Model explainability

- The banks has to justify the decision of approving or not approving the credit
- We calculate the most important features for each prediction separately

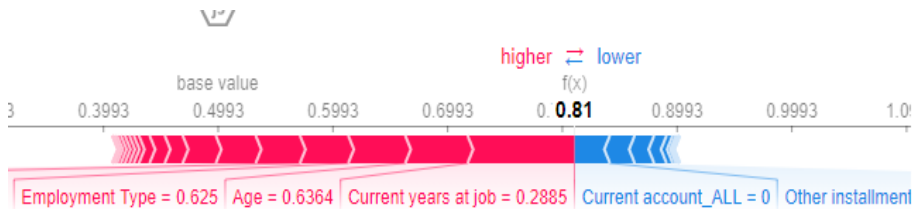


Figure: Local feature importance



Dataset exploration

The goal of the second part was to develop a general credit assessment tool

We explored a few datasets :

- UCI Repository : Multiple credit datasets from various countries, only a small number of usable features
- PAKDD 2010 : Too few usable features
- Home Credit Default Risk : Unusable in our context
- Fast Credit : Too few samples

Lending Club dataset

Huge number of samples ($> 1M$), sufficient number of usable features, the only dataset we found really usable for our purpose.



Model generalization

- We finally selected 13 features from the generalization Dataset for the final model analysis.
- From 3 different aspects :
 - Credit information required by the client
 - Client's Personal information
 - Traceability of the past credit demanded by the client
- Balanced Random Forest Model with $AUC = 0.64$



Web application functionalities

- Require all the necessary information from the client using a multi step form
- Output the probability that the client gets an approbation for the credit

Credit Simulation

1

 Credit Information

crf

 Client information

crf

 Credit history

4

 Done

How much do you want to borrow? *

With what interest (negotiable) ? *

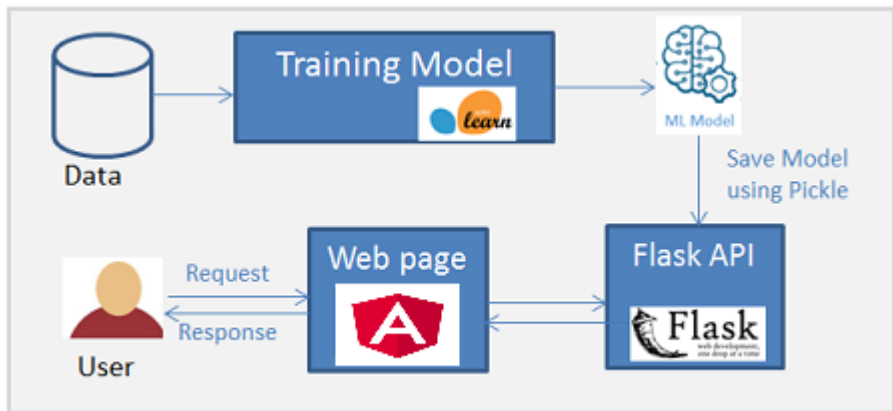
With what duration? *

Is it an individual application or a joint application with co-borrowers? *

Next



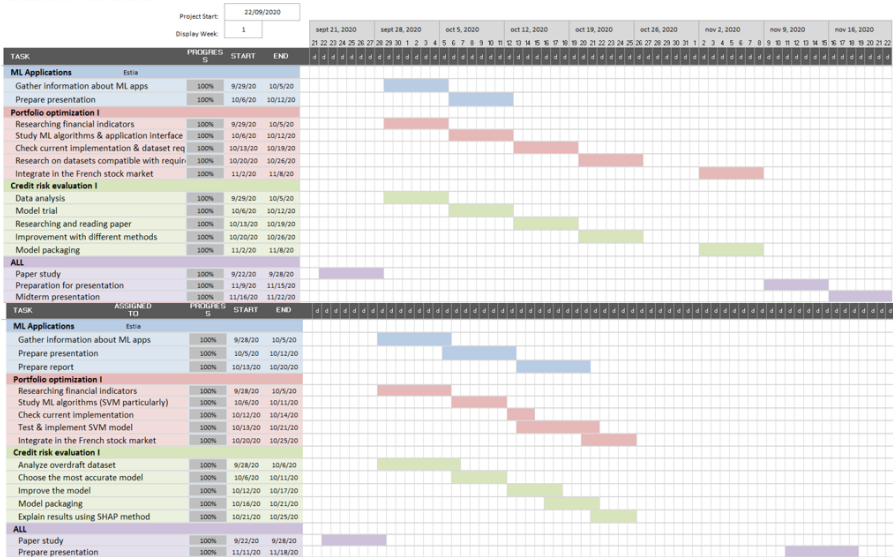
Web application architecture



Project before & after

Comparing Gantt charts I

What use of AI in finance ?



Comparing Gantt charts II

What use of AI in finance ?

Project Start: 22/09/2020
Display Week: 9

nov 16, 2020 nov 23, 2020 nov 30, 2020 déc 7, 2020 déc 14, 2020 déc 21, 2020 déc 28, 2020 janv 4, 2021
16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 1 2 3 4 5 6 7 8 9 10



State of the project

Portfolio Optimization

- Start mostly from scratch
- Based on Haifei Zhang's Master Thesis
- Adapting to CAC40 dataset
- Improving SVM model
- Adding Deep Learning methods (LSTM)
- Make 7-day predictions

Credit Risk

- Start mostly from scratch
- Based on Zinan Zhou's Master Thesis
- Implement Balanced Random Forest for Overdraft
- Adapt to Lending club dataset
- Build a web application

Conclusions

- The study of AI applications helped to create an idea of how the AI and finance world are connected as well as how our two applications are placed.
- For the portfolio management problem, we studied previous and state-of-the-art algorithms and implemented for the French stock market dataset. We used different models of SVM and LSTM.
- For the credit risk assessment problem, we analysed the loan data and after implementing various methods we concluded that the best method to use is **Balanced Random Forest**. We chose to implement it to the Lending club dataset and built a web application.

